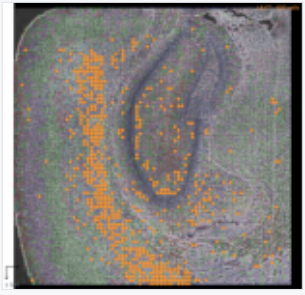
Fig. 2 From a tissue section to the genome folding of its cells — and the same interaction on every kind of data

a The workflow, end to end (browser ⇅ Python, data from the atlas)

1. open the brain section from the atlas (HTTP)
2. spot compartments on the tissue
3. spot composition from its RNA
4. cell-type compartments from single-cell Hi-C
5. expected vs observed; residual on the tissue
6. replicate: section 2, 10-µm spatial Hi-C

b Compartments of each spot

browser: per-spot A/B on the section



c What each spot is made of

RNA → cell-type fractions (API)

maps of L2/3, L6, Oligo fractions on the section

d Expected from the cell types

Σ fraction × cell-type A/B
(GAGE-seq, Paired Hi-C, scHiCAR)

observed vs expected (r, null)

e What the cell types do not explain

browser: residual per spot on the section

residual map (outcome-dependent)

f Replicated

residual vs cortical depth:
section 1, section 2, 10-µm cortex

depth profile × 3 datasets (outcome-dependent)

g One locus, three views

maps of deep vs superficial spots;
the locus in HiRES 3-D structures

example locus (outcome-dependent)

h The same interaction on the other kinds of data

select a cell type; every view follows (browser captures)

single-cell Hi-C + RNA (HiRES brain): Oli



chromatin tracing + IF (Takei cerebellum): Bergmann glia



orange dashed: content depends on the feasibility analysis (residual present → position-dependent folding; absent → cell types explain folding in situ)